

Smart Expense Tracker — Database Design

1. Functionality

The Smart Expense Tracker is designed to manage and track the user's financial transactions.

The system will provide the following functionalities:

- User details
- Bank account and card details
- Income and expense transactions
- Expense categories
- Payment modes such as UPI and Card
- Monthly category-wise budgets
- Budget usage and remaining amount
- Budget status such as Safe, Near Limit and Exceeded
- Financial transaction reports

2. Entities

The following entities are identified for the system:

1. User
2. Account
3. Category
4. Transaction
5. Budget

According to the reference format, each entity will correspond to a database table, and its attributes will become columns in that table.

2.1 User

Attributes:

- `user_id` — Primary Key
- `name`
- `email` — Unique
- `phone_number` — Unique

Relationship:

One user can have multiple accounts and multiple budgets.

3. Attributes and Relationships

3.1 User

Attributes:

- `user_id` (Primary Key)
- `name`
- `email` (Unique)
- `phone_number` (Unique)

Relationships:

- One User can have many Accounts (**One-to-Many**).
 - One User can have many Budgets (**One-to-Many**).
-

3.2 Account

Attributes:

- `account_id` (Primary Key)
- `user_id` (Foreign Key)
- `bank_name`
- `account_type`
- `last_digits`

Relationships:

- Many Accounts belong to one User (**Many-to-One**).
 - One Account can have many Transactions (**One-to-Many**).
-

3.3 Category

Attributes:

- `category_id` (Primary Key)
- `category_name`
- `category_type`

Relationships:

- One Category can have many Transactions (**One-to-Many**).
 - One Category can have many Budgets (**One-to-Many**).
-

3.4 Transaction

Attributes:

- `transaction_id` (Primary Key)
- `account_id` (Foreign Key)
- `category_id` (Foreign Key)
- `amount`
- `merchant_name`
- `payment_mode`
- `transaction_date`
- `transaction_type`

Relationships:

- Many Transactions belong to one Account (**Many-to-One**).
 - Many Transactions belong to one Category (**Many-to-One**).
-

3.5 Budget

Attributes:

- `budget_id` (Primary Key)
- `user_id` (Foreign Key)
- `category_id` (Foreign Key)

- budget_amount
- month
- year

Relationships:

- Many Budgets belong to one User (**Many-to-One**).
- Many Budgets belong to one Category (**Many-to-One**).

4. Table Structure

4.1 Users

Table Name: `users`

Purpose: Stores the basic details of the application users.

```
mysql> desc users;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra          |
+-----+-----+-----+-----+-----+-----+
| user_id        | int           | NO   | PRI | NULL    | auto_increment |
| name           | varchar(100)  | NO   |     | NULL    |                |
| email          | varchar(100)  | YES  | UNI | NULL    |                |
| phone_number   | varchar(15)   | YES  | UNI | NULL    |                |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

4.2 Categories

Table Name: `categories`

Purpose: Stores the categories used to classify income and expenses.

```
mysql> desc categories;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra          |
+-----+-----+-----+-----+-----+-----+
| category_id    | int           | NO   | PRI | NULL    | auto_increment |
| category_name   | varchar(50)   | NO   |     | NULL    |                |
| category_type   | varchar(20)   | NO   |     | NULL    |                |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

4.3 Accounts

Table Name: `accounts`

Purpose: Stores the bank accounts and cards associated with a user.

```
mysql> desc accounts;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra          |
+-----+-----+-----+-----+-----+-----+
| account_id     | int           | NO   | PRI | NULL    | auto_increment |
| user_id        | int           | NO   | MUL | NULL    |                |
| bank_name      | varchar(100)  | NO   |     | NULL    |                |
| account_type   | varchar(30)   | NO   |     | NULL    |                |
| last_digits    | varchar(4)    | NO   |     | NULL    |                |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

4.4 Transactions

Table Name: `transactions`

Purpose: Stores individual income and expense transactions.

```
mysql> desc transactions;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra          |
+-----+-----+-----+-----+-----+-----+
| transaction_id | int           | NO   | PRI | NULL    | auto_increment |
| account_id     | int           | NO   | MUL | NULL    |                |
| category_id    | int           | NO   | MUL | NULL    |                |
| amount         | decimal(10,2) | NO   |     | NULL    |                |
| merchant_name  | varchar(100)  | NO   |     | NULL    |                |
| payment_mode   | varchar(30)   | NO   |     | NULL    |                |
| transaction_date | datetime      | NO   |     | NULL    |                |
| transaction_type | varchar(20)   | NO   |     | NULL    |                |
+-----+-----+-----+-----+-----+-----+
8 rows in set (0.00 sec)
```

4.5 Budgets

Table Name: `budgets`

Purpose: Stores monthly spending limits for different categories.

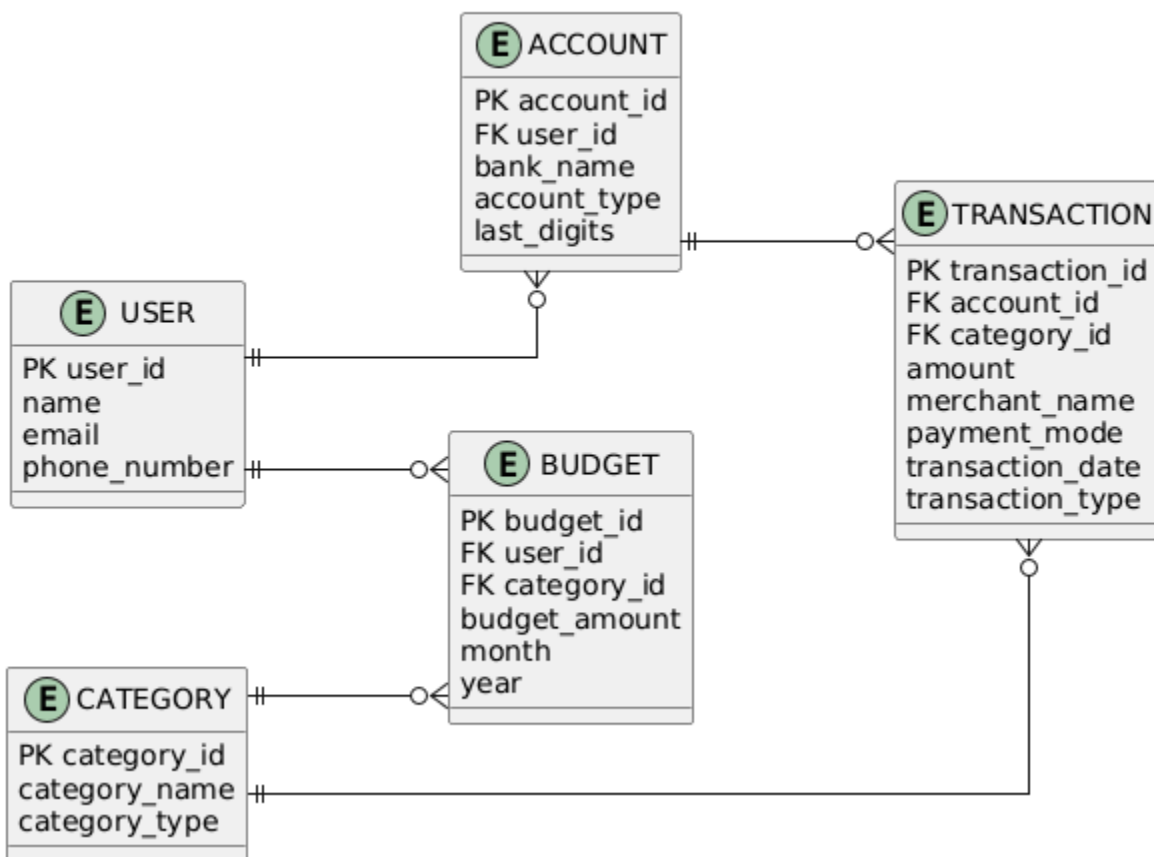
```
mysql> desc budgets;
```

Field	Type	Null	Key	Default	Extra
budget_id	int	NO	PRI	NULL	auto_increment
user_id	int	NO	MUL	NULL	
category_id	int	NO	MUL	NULL	
budget_amount	decimal(10,2)	NO		NULL	
month	int	NO		NULL	
year	int	NO		NULL	

```
6 rows in set (0.00 sec)
```

5. ER Diagram

Smart Expense Tracker - ER Diagram



5.1 Relationship Description

The ER diagram represents the relationships between the entities of the Smart Expense Tracker database:

1. **User – Account:** One user can have multiple accounts.
2. **User – Budget:** One user can create multiple budgets.
3. **Account – Transaction:** One account can have multiple transactions.
4. **Category – Transaction:** One category can be associated with multiple transactions.
5. **Category – Budget:** One category can have multiple budgets.

5.2 Cardinality

The relationships in the ER diagram are **one-to-many (1:M)** relationships:

Relationship	Cardinality
User → Account	1 : M
User → Budget	1 : M
Account → Transaction	1 : M
Category → Transaction	1 : M
Category → Budget	1 : M

6. Creating Database and Tables

The Smart Expense Tracker database is created using **MySQL**. The database is named `expense_tracker`.

6.1 Create Database

```
CREATE DATABASE expense_tracker;
```

The above command creates a new database named **expense_tracker**.

6.2 Select Database

Before creating tables, the database must be selected using the `USE` statement.

```
USE expense_tracker;
```

The above command selects the `expense_tracker` database for further operations.

6.3 Create Users Table

```
CREATE TABLE users (  
    user_id INT PRIMARY KEY AUTO_INCREMENT,  
    name VARCHAR(100) NOT NULL,  
    email VARCHAR(100) UNIQUE,  
    phone_number VARCHAR(15) UNIQUE  
);
```

Purpose: Stores the basic details of users.

6.4 Create Categories Table

```
CREATE TABLE categories (
```



```
category_id INT PRIMARY KEY AUTO_INCREMENT,  
category_name VARCHAR(50) NOT NULL,  
category_type VARCHAR(20) NOT NULL  
);
```

Purpose: Stores categories used to classify income and expenses.

6.5 Create Accounts Table

```
CREATE TABLE accounts (  
    account_id INT PRIMARY KEY AUTO_INCREMENT,  
    user_id INT NOT NULL,  
    bank_name VARCHAR(100) NOT NULL,  
    account_type VARCHAR(30) NOT NULL,  
    last_digits VARCHAR(4) NOT NULL,  
    FOREIGN KEY (user_id) REFERENCES users(user_id)  
);
```

Purpose: Stores the bank accounts and cards associated with users.

6.6 Create Transactions Table

```
CREATE TABLE transactions (  
    transaction_id INT PRIMARY KEY AUTO_INCREMENT,  
    account_id INT NOT NULL,  
    category_id INT NOT NULL,  
    amount DECIMAL(10,2) NOT NULL,
```

```
merchant_name VARCHAR(100) NOT NULL,  
payment_mode VARCHAR(30) NOT NULL,  
transaction_date DATETIME NOT NULL,  
transaction_type VARCHAR(20) NOT NULL,  
FOREIGN KEY (account_id) REFERENCES accounts(account_id),  
FOREIGN KEY (category_id) REFERENCES categories(category_id)  
);
```

Purpose: Stores individual income and expense transactions.

6.7 Create Budgets Table

```
CREATE TABLE budgets (  
    budget_id INT PRIMARY KEY AUTO_INCREMENT,  
    user_id INT NOT NULL,  
    category_id INT NOT NULL,  
    budget_amount DECIMAL(10,2) NOT NULL,  
    month INT NOT NULL,  
    year INT NOT NULL,  
    FOREIGN KEY (user_id) REFERENCES users(user_id),  
    FOREIGN KEY (category_id) REFERENCES categories(category_id)  
);
```

Purpose: Stores monthly spending limits for different categories.